

RAHUL SHARMA

500 Bartholomew Rd, Piscataway, NJ 08854-8058

✉ rhsharma8272@gmail.com

☎ +1 463-701-7908

🔗 [LinkedIn](#) | [Google scholar](#) | [GitHub](#)

EDUCATION

Rutgers, The State University of New Jersey Ph.D. in Mechanical Engineering	Sep 2021 - May 2026 GPA – 3.98/4.0
Indian Institute of Technology (IIT), Kanpur, India M.Tech in Mechanical Engineering	Dec 2015 - Dec 2017 GPA – 8.75/10
Punjab Technical University, India B.Tech in Mechanical Engineering	Aug 2011 - May 2015 GPA – 72.6/100

SKILLS

Programming: Python, MATLAB, Fortran	Tools: Anaconda, Jupyter, Wandb, Linux
AI/ML: PyTorch, Scikit-learn, Git, Docker	Simulation: COMSOL, Abaqus, Netfabb, GT-suite
Familiar with: TensorFlow, Keras, CI/CD	Design: SolidWorks, Creo
Data Processing: Pandas, NumPy, SciPy, Matplotlib	

RESEARCH/WORK EXPERIENCE

Research Assistant	Rutgers University, New Jersey	<i>Sep 2021 - May 2026</i>
- Built physics-informed machine learning (PINN) models using PyTorch that accurately predicted system temperature and stress distribution with ~99.8% accuracy. - Created comprehensive physics-based simulation of metal additive manufacturing processes using commercial software COMSOL Multiphysics and validated with experimental data. - Integrated ML model with simulation software, cutting runtime by ~80%. - Developed end-to-end testing pipeline with CI/CD practices, implementing comprehensive automated PDE-residual verification, boundary-condition enforcement, and numerical-stability checks. - Developed an LSTM-based deep learning model to predict fatigue failure of 3D-printed components. - Led cross-functional teams to develop a 5G-enabled Industry 4.0 testbed for real-time monitoring and analysis of the milling process. - Hands-on expertise with AconityMINI metal 3D-printer, including print preparation using Netfabb (support generation, hatching, slicing), setup, and monitoring via integrated pyrometers/CMOS camera - Published 10+ papers in high-impact top-tier conferences and journals on cutting-edge advancements in AI and smart manufacturing.		
ML Consultant (Intern)	Intelligent Twins, California	<i>Sep 2025 - present</i>
- Worked on Swin Transformer model with physics-informed loss functions to predict temperature and thermal warpage in microchip for Vinci4d.ai (startup) - Integrated ML predictions with physics-based solvers by writing user-defined Autograd functions to provide improved initial guess and accelerate simulation convergence. - Developed unit tests to benchmark loss formulations, numerical preconditioners, multigrid solvers, and training-strategy selection for CNN-based physics-informed model. - Leveraged Docker containers to standardize and reproduce large-scale physics-based ML training and testing workflows on Google Cloud GPU infrastructure. - Gain hands-on experience collaborating with other ML engineers on GitHub while working on complete machine learning development pipeline.		

- Evaluate the structural integrity of the engine-mounted components by performing the modal analysis, correlating modes with experimental data, and applying dynamic/G-load using **Abaqus**.
- Studied and optimized the effect of valve train parameters on contact stress between cam and roller using **GT-Suite** (1D simulation) and **Creo Simulate** (3D simulation).
- Led a team of 5 in completing **20+ projects** over three years, overseeing data pre-processing, analysis, and post-processing to deliver actionable solutions.
- Automated repetitive pre-processing workflows by developing custom **Python** scripts, significantly reducing manual effort and improving simulation turnaround time

- Lecturer of the course Smart Manufacturing, focusing on advanced manufacturing processes, data ingestion, data integration, physics/data-driven models, model-based predictive controls, and industrial practices in different manufacturing sectors.

PUBLICATIONS

Journal Papers

- **R. Sharma**, Y.B. Guo, Physics-Informed Machine Learning for Melt Pool Dynamics in Laser Powder Bed Fusion, 2025 (submitted)
- **R. Sharma**, M. Raissi, Y.B. Guo, Physics-Informed Machine Learning Regulated by Finite Element Analysis for Simulation Acceleration of Laser Powder Bed Fusion, Computation Mechanics (in review), 2025. ([Link](#))
- W. Liu, **R. Sharma**, Y.B. Guo, AI-Driven Digital Twins for Smart Milling: A Comprehensive Review. Engineering, 2025. ([Link](#))
- **R. Sharma**, Y.B. Guo, Thermo-mechanical physics-informed deep learning for prediction of thermal stress evolution in laser metal deposition, Engineering Applications of AI, 2025. ([Link](#))
- D. Patel, **R. Sharma**, Y.B. Guo, Computational, Data-Driven, and Physics-Informed Machine Learning Approaches for Microstructure Modeling in Metal AM. Ann Rev of HT, 2025. ([Link](#))
- **R. Sharma**, M. Raissi, G. Guo, Y.B. Guo, 2023, "Physics-Informed Machine Learning of Argon Gas-Driven Melt Pool Dynamics," ASME. J. Manuf. Sci. Eng. 146(8): 081008, 2024. ([Link](#))
- **R. Sharma**, M. Raissi, Y.B. Guo, Physics-informed deep learning of gas flow-melt pool multi-physical dynamics during powder bed fusion, CIRP Annals, 72/1: 161-164, 2023. ([Link](#))
- **R. Sharma**, Y.B. Guo, Computational Modeling and Physics-Informed Machine Learning of Metal Additive Manufacturing: State-of-the-Art and Future Perspective. Annual Review of Heat Transfer, 2022. ([Link](#))
- **R. Sharma**, A. Kumar, Track-Scale Simulations of Selective Laser Melting to Investigate Development and Mitigation of Thermal Stresses. Lasers Manuf. Mater. Process. 6: 464-492, 2019. ([Link](#))
- S. Saxena, **R. Sharma**, A. Kumar, A Microscale Study of Thermal Field, and Stresses during Processing of Ti6Al4V Powder Layer by Selective Laser Melting. Lasers Manuf. Mater. Process., 5: 335-365, 2018. ([Link](#))

Conference papers

- **R. Sharma**, Y.B. Guo, 2025, "Physics-Informed Machine Learning of Thermal Stress Evolution in Laser Metal Deposition", In: TMS 2025 154th Annual Meeting & Exhibition Supplemental Proceedings. TMS 2025. The MMM Series. Springer, Cham. ([Link](#))
- **R. Sharma**, M. Raissi, Y.B. Guo, 2024, Physics-Informed Machine Learning for Smart Additive Manufacturing, 18th CIRP Conference on Intelligent Computation in Manufacturing Engineering - CIRP ICME '24, 10-12 July, Procedia CIRP, Elsevier, ISSN: 2212-8271. ([Link](#))

Book Chapters

- **Sharma R.**, Saxena S., Kumar A. (2019) Thermal Stress Analysis in Selective Laser Melting of Ti6Al4V Powder Layer, In: Dixit U., Joshi S., Davim J. (eds) App. of Lasers in Manuf., Springer, Singapore. ([Link](#))
- **R. Sharma**, Y.B. Guo, R.X. Gao (2025) Residual Stress Induced by Grinding, Brett A. Miller, Roch J. Shipley, Ronald J. Parrington, Daniel P. Dennies, (eds) ASM Handbook, Volume 25A, Residual Stress Fundamentals. ([Link](#))